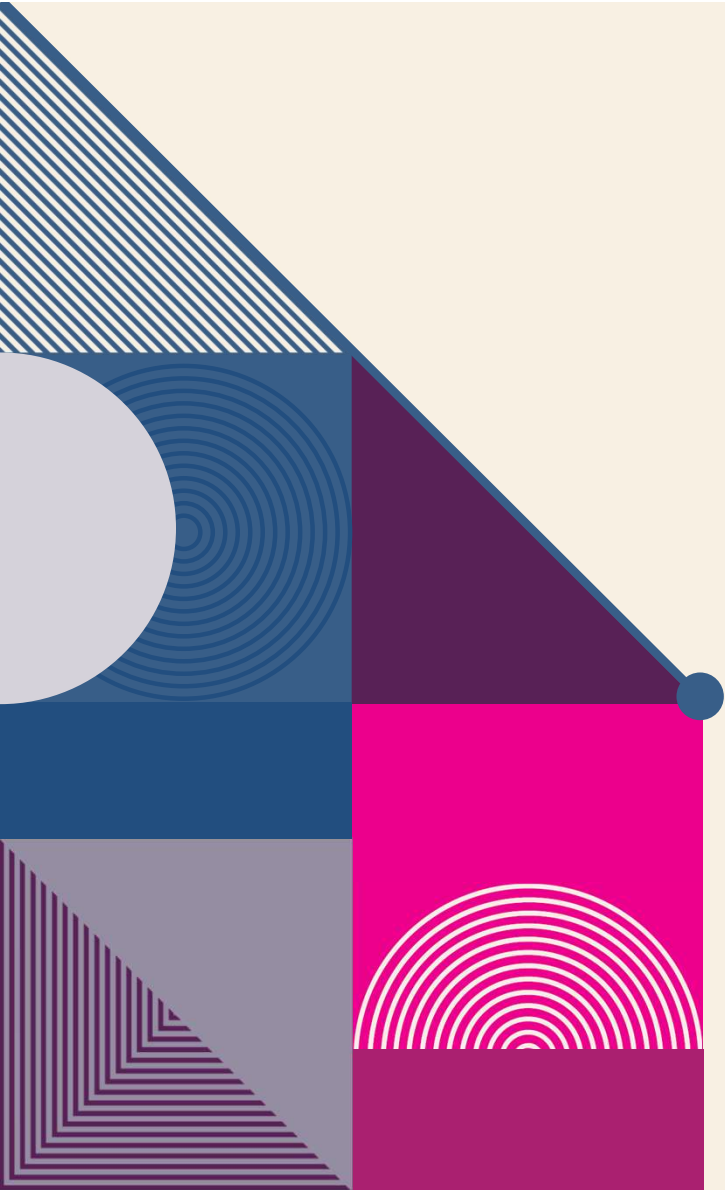


**IOT-ENABLED SMART
SOLUTIONS FOR
SUSTAINABLE
AIRCRAFT INTERIORS**



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THEME :

**ADVANCED SUSTAINABLE DESIGN FOR
AIRCRAFT INTERIORS**



PROBLEM STATEMENT:

Embraer is committed to designing sustainable aircraft interiors that reduce environmental impact without compromising aesthetics and comfort. The main challenges include sourcing eco friendly materials ensuring durability and maintaining cost effectiveness the goal is to create an aircraft interior design that is both sustainable and appealing to passengers.

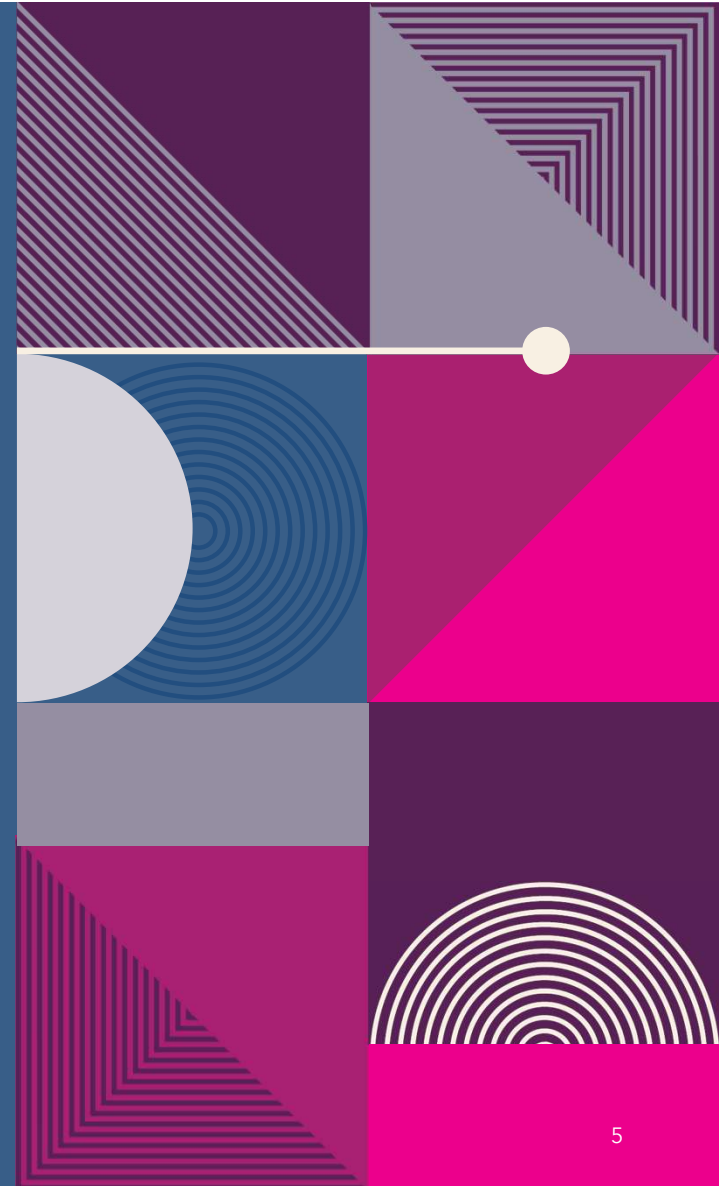
SOLUTION:

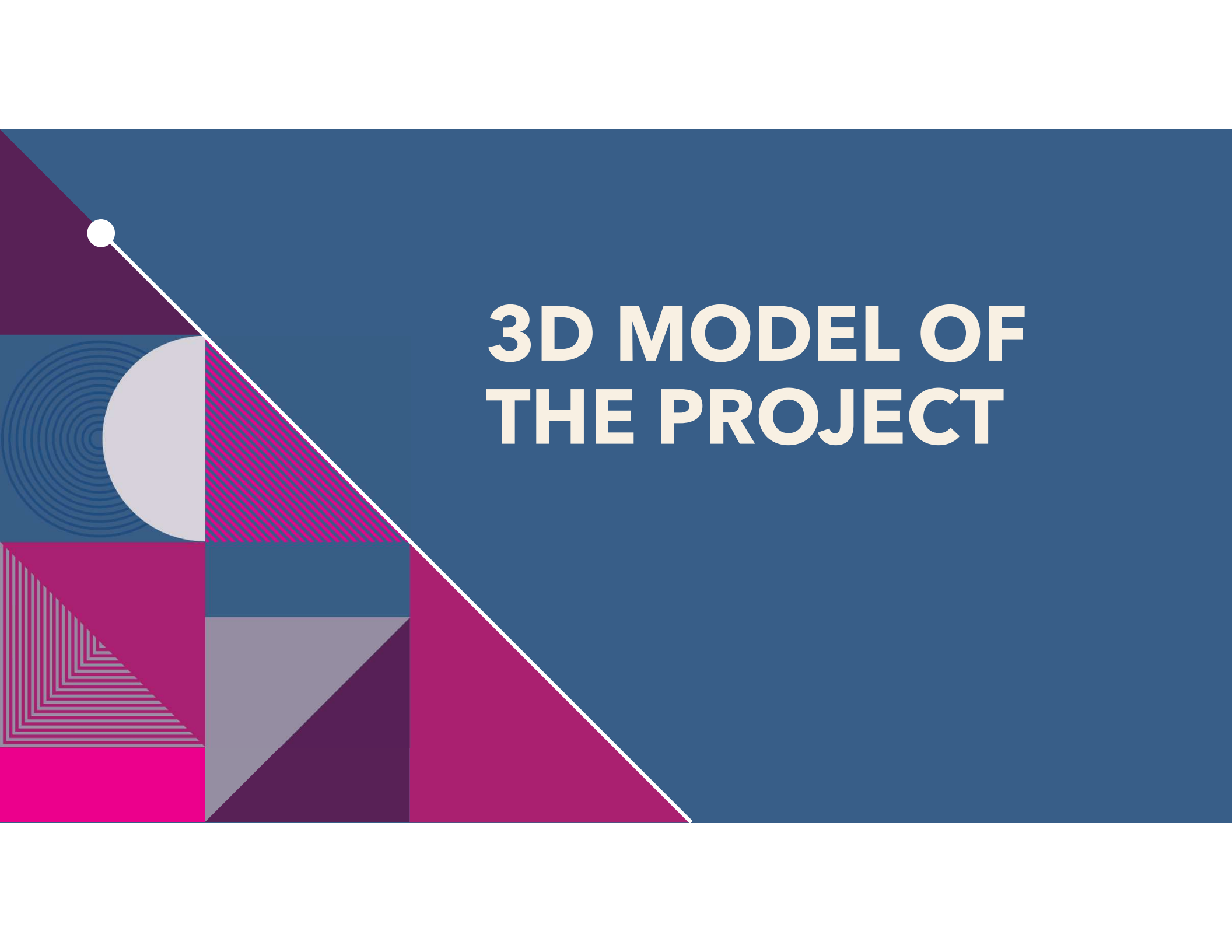
To design sustainable aircraft interiors, Embraer can focus on using eco-friendly materials such as recycled composites, biodegradable fabrics, and bio-based foams for seats and panels.

Adopting a modular design approach will allow for easy replacement and repair of interior components, reducing waste and extending the lifespan of parts. Incorporating lightweight materials and structures can decrease the aircraft's overall weight, improving fuel efficiency and reducing emissions.

The use of 3D printing technology can facilitate on-demand production of interior parts, minimizing waste and enabling customization. Additionally, integrating energy-efficient systems, like LED lighting and smart cabin technologies, can reduce energy consumption.

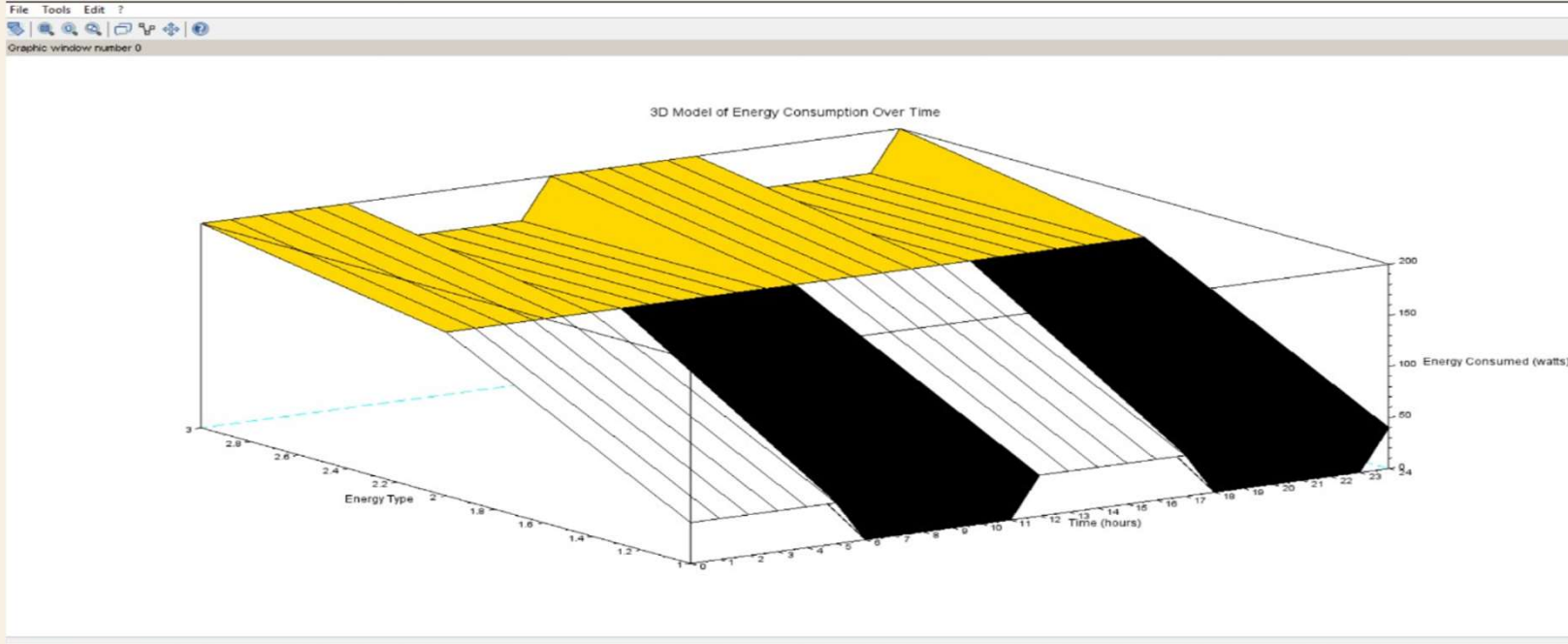
Finally, designing interiors for easy disassembly and recycling will ensure that materials can be reused at the end of their lifecycle, promoting a circular economy approach.



The image features a dark blue background. A diagonal line from the top-left to the bottom-right divides the space. The area to the left of this line is filled with various geometric patterns: a dark purple triangle at the top-left, a light grey semi-circle, concentric blue circles, a pink hatched triangle, a pink square with a white line pattern, a grey triangle, and a pink square. A small white circle is located on the diagonal line near the top-left. The text '3D MODEL OF THE PROJECT' is written in a bold, white, sans-serif font on the right side of the image.

3D MODEL OF THE PROJECT

3D MODEL OF ENERGY CONSUMPTION OVER TIME



CODE:

```
UNSTOP.sci (C:\Users\tanuv\Documents\UNSTOP.sci) - SciNotes
File Edit Format Options Window Execute ?
UNSTOP.sci (C:\Users\tanuv\Documents\UNSTOP.sci) - SciNotes
*UNSTOP.sci X

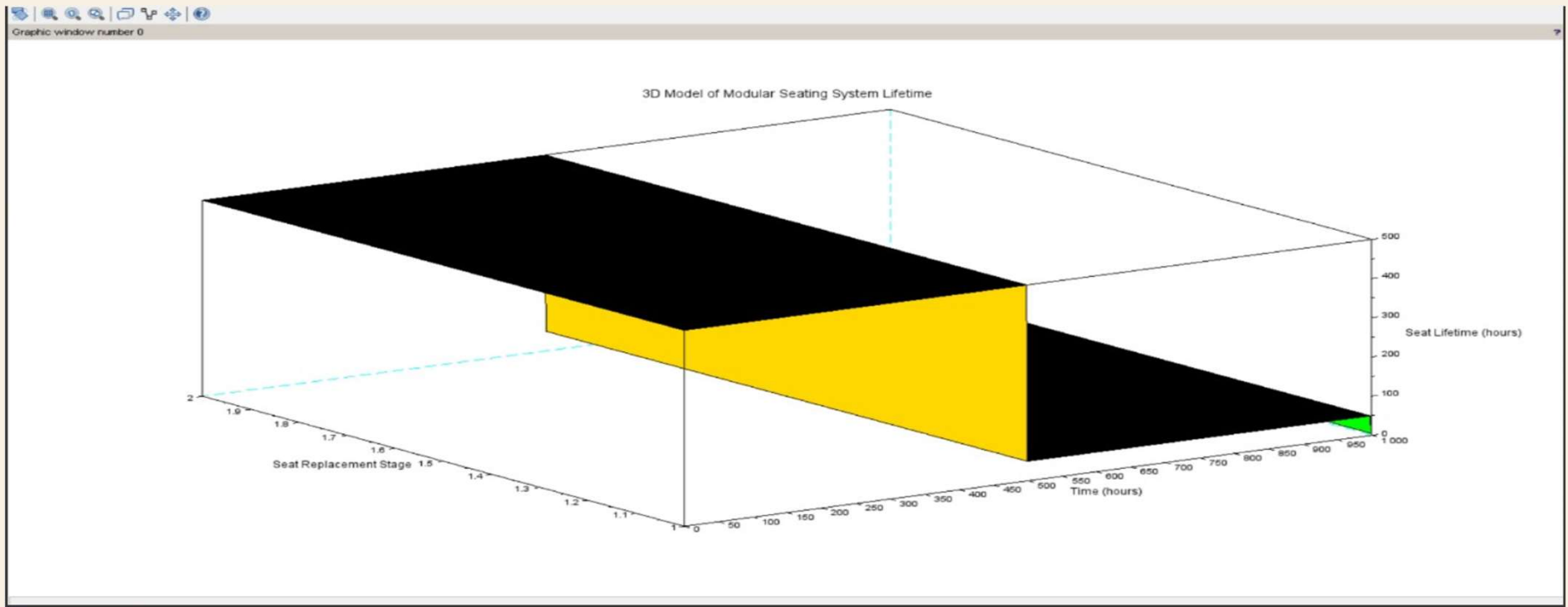
1 // Define parameters
2 power_lights = 50; // Power consumption for lighting in watts
3 power_ac = 200; // Power consumption for air conditioning in watts
4 efficiency_factor = 0.8; // Efficiency factor for energy-saving systems
5
6 // Time parameters
7 time = 0:1:24; // Time in hours
8 n = length(time); // Number of time steps
9
10 // Initialize arrays
11 energy_consumed = zeros(1, n);
12 energy_ac = zeros(1, n);
13 energy_lights = zeros(1, n);
14
15 // Simulation loop
16 for i = 1:n
17     if modulo(time(i), 12) < 6 then // Use modulo instead of mod
18         energy_ac(i) = power_ac * efficiency_factor;
19         energy_lights(i) = power_lights * efficiency_factor;
20         energy_consumed(i) = (power_lights + power_ac) * efficiency_factor;
21     else
22         energy_ac(i) = power_ac * efficiency_factor;
23         energy_lights(i) = 0; // No lights after 6 hours
24         energy_consumed(i) = power_ac * efficiency_factor;
25     end
26 end
27
28 // Create a mesh grid for 3D plot
29 [Time, Energy] = meshgrid(time, [1:2:3]); // Create grid for 3 different energy types
30
31 // Plot the 3D model
32 clf;
33 surf(Time, Energy, [energy_lights; energy_ac; energy_consumed]);
34 title('3D Model of Energy Consumption Over Time');
35 xlabel('Time (hours)');
36 ylabel('Energy Type');
37 zlabel('Energy Consumed (watts)');
```




EXPLANATION:

- The code simulates energy consumption for lighting and air conditioning (AC) over 24 hours. Lighting consumes 50 watts, and AC consumes 200 watts, both with an efficiency factor of 0.8. The day is split into 6-hour periods:
- during the first 6 hours (0-5, 12-17), both lighting and AC are active; during the next 6 hours (6-11, 18-23), only the AC is used. Arrays store energy values for each hour .
- A 3D surface plot is generated to show how energy consumption varies over time. The x-axis represents time, the y-axis shows energy types (lighting, AC, combined), and the z-axis displays energy consumption in watts. The plot visually highlights energy usage changes throughout the day.

3D MODEL OF MODULAR SEATING SYSTEM LIFETIME



CODE:

```
UNSTOP.sci (C:\Users\tanuv\Documents\UNSTOP.sci) - SciNotes
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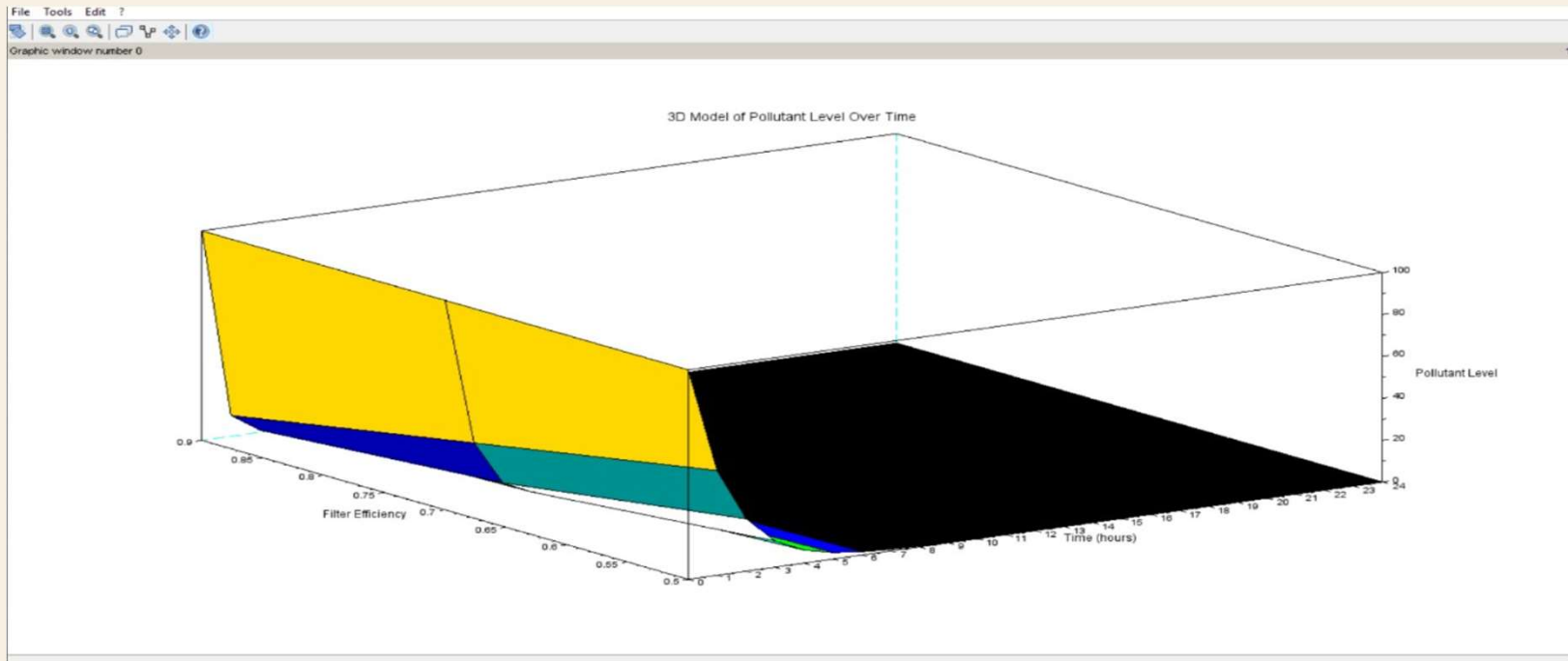
1 // Define parameters
2 initial_seat_lifetime = 500; // Lifetime of each seat module in hours
3 replacement_rate = 0.1; // Rate at which seats are replaced
4
5 // Time parameters
6 time = 0:1:1000; // Time in hours
7 n = length(time); // Number of time steps
8
9 // Initialize array
10 seat_lifetime = initial_seat_lifetime * ones(1, n);
11
12 // Simulation loop
13 for i = 2:n
14     if modulo(i, initial_seat_lifetime) == 0 then // Use modulo to simulate seat replacement
15         seat_lifetime(i) = seat_lifetime(i-1) * replacement_rate; // Reduce the lifetime
16     else
17         seat_lifetime(i) = seat_lifetime(i-1); // Lifetime remains unchanged
18     end
19 end
20
21 // Create 3D mesh grid
22 [Time, Lifetime] = meshgrid(time, [1 2]); // Represent the grid for seat lifetime over time
23
24 // Plot the 3D model
25 clf;
26 surf(Time, Lifetime, [seat_lifetime; seat_lifetime]); // Plotting 3D surface
27 title('3D Model of Modular Seating System Lifetime');
28 xlabel('Time (hours)');
29 ylabel('Seat Replacement Stage');
30 zlabel('Seat Lifetime (hours)');
```



EXPLANATION:

- The Scilab code simulates the lifetime of modular seating over 1000 hours, adjusting for periodic replacements every 500 hours. Initially, each seat lasts 500 hours, but after each replacement, its lifetime is reduced to 10% of the previous value. The code generates a 3D plot where the x-axis represents time, the y-axis represents replacement stages (initial and post-replacement), and the z-axis shows the seat lifetime in hours. This visualization illustrates how the seat lifetime evolves over time, incorporating the effects of replacements.

3D MODEL OF POLLUTANT LEVEL OVER TIME



CODE:

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UNSTOP.sci

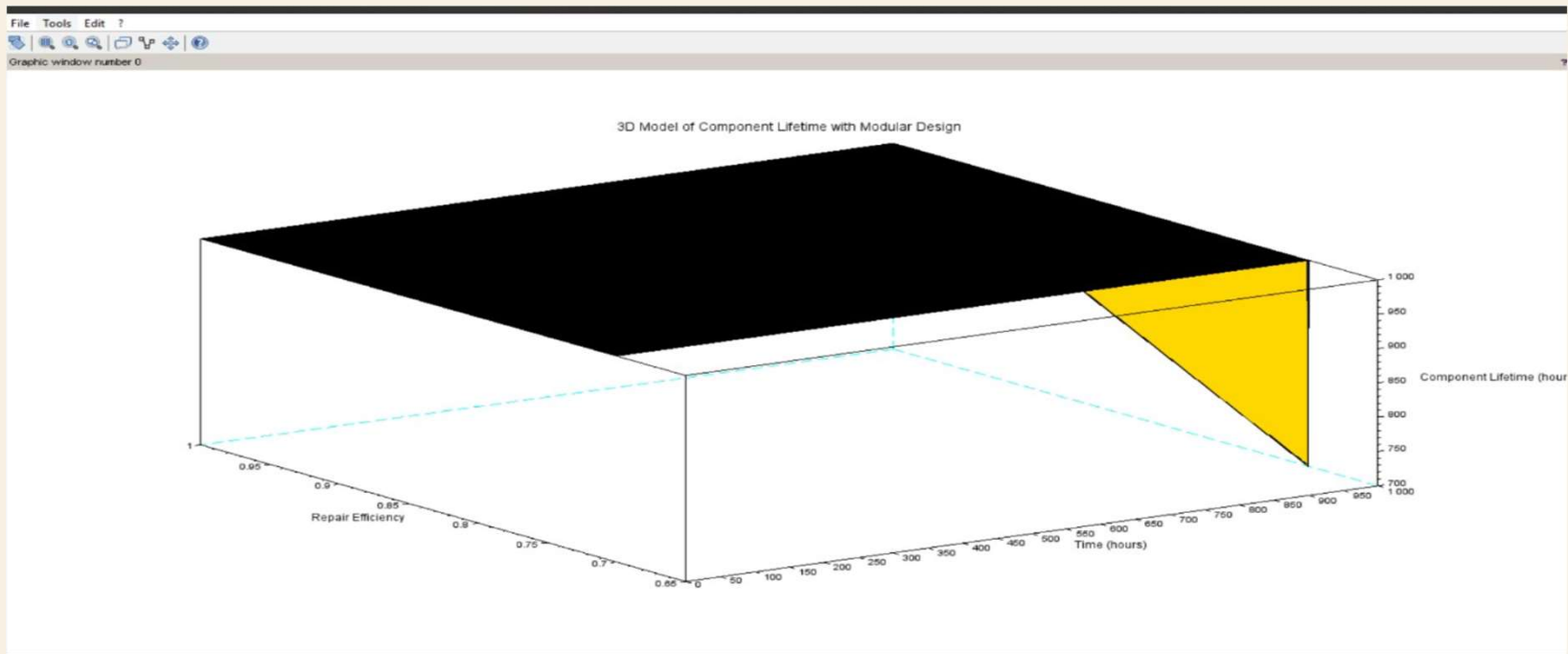
```
1 //Define parameters
2 initial_pollutant_level = 100; //Initial pollutant level
3 filter_efficiencies = [0.5, 0.7, 0.9]; //Different efficiencies of filters
4
5 //Time parameters
6 time = 0:1:24; //Time in hours
7 n = length(time); //Number of time steps
8 m = length(filter_efficiencies); //Number of filter efficiency scenarios
9
10 //Initialize array
11 pollutant_level = zeros(m, n); //Array to hold pollutant levels for each efficiency
12
13 //Simulation loop
14 for j = 1:m
15     filter_efficiency = filter_efficiencies(j);
16     for i = 1:n
17         if i == 1 then
18             pollutant_level(j, i) = initial_pollutant_level;
19         else
20             pollutant_level(j, i) = pollutant_level(j, i-1) * (1 - filter_efficiency); //Decrease pollutant level
21         end
22     end
23 end
24
25 //Create 3D mesh grid
26 [Time, Efficiency] = meshgrid(time, filter_efficiencies); //Create grid for time and filter efficiencies
27
28 //Plot the 3D model
29 clf;
30 surf(Time, Efficiency, pollutant_level); //Plotting 3D surface
31 title('3D Model of Pollutant Level Over Time');
32 xlabel('Time (hours)');
33 ylabel('Filter Efficiency');
34 zlabel('Pollutant Level');
```




EXPLANATION:

- The Scilab code models how pollutant levels change over a 24-hour period for various filter efficiencies (0.5, 0.7, and 0.9). It calculates pollutant levels at each hour by applying a reduction factor based on the filter's efficiency. The data is then visualized in a 3D plot, where the x-axis represents time, the y-axis represents different filter efficiencies, and the z-axis shows the pollutant levels. This 3D surface plot provides a clear view of how different filter efficiencies affect the rate of pollutant removal over time.

COMPONENT LIFETIME WITH MODULAR DESIGN



CODE:

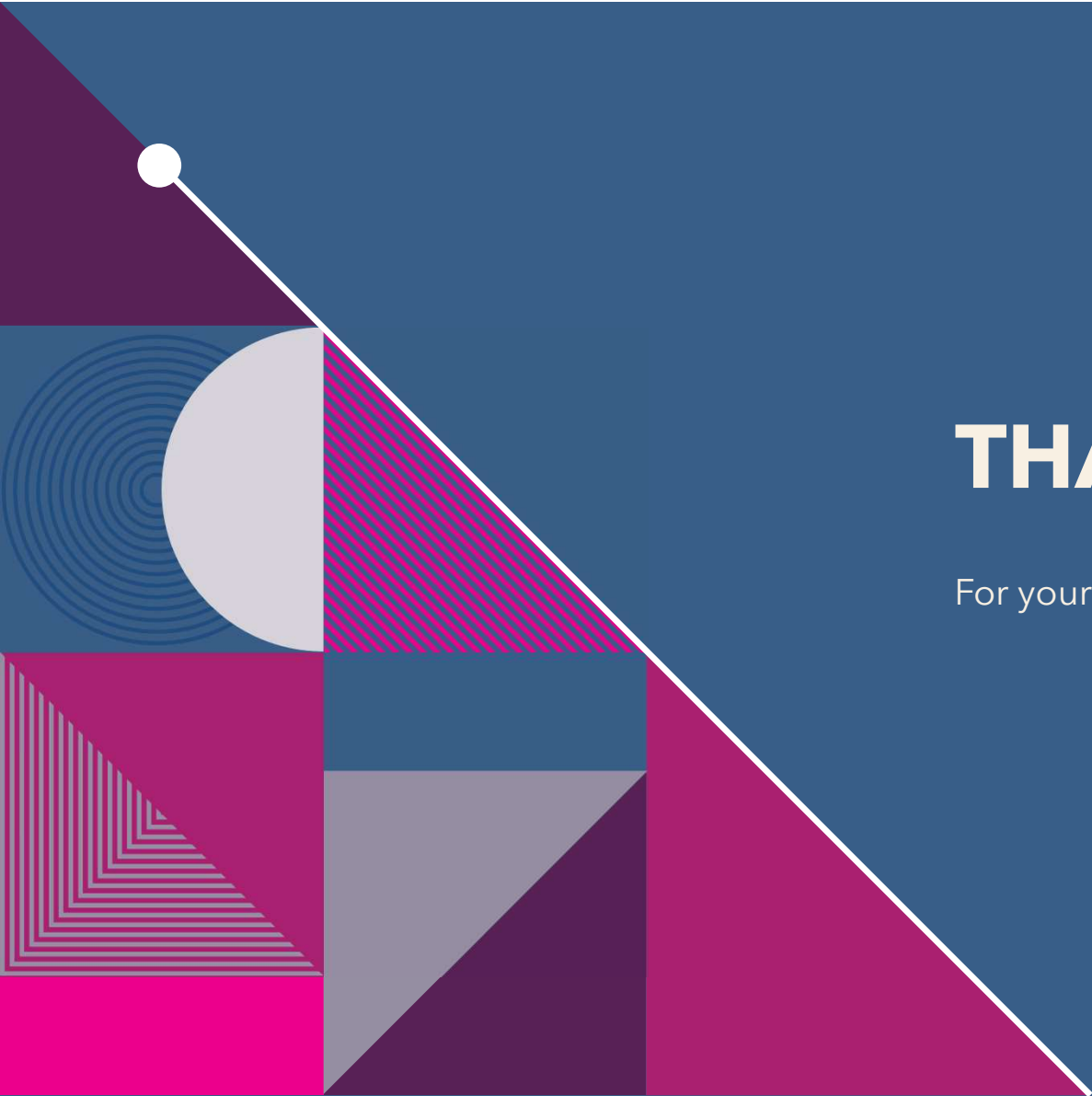
```
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UNSTOP.sci (C:\Users\tanuv\Documents\UNSTOP.sci) - SciNotes

*UNSTOP.sci X
1  Define parameters
2  initial_component_lifetime = 1000; /// Lifetime of each component
3  repair_efficiencies = [0.7, 0.9, 1.0]; /// Different repair efficiencies
4
5  Time parameters
6  time = 0:1:1000; /// Time in hours
7  n = length(time); /// Number of time steps
8  m = length(repair_efficiencies); /// Number of repair efficiency scenarios
9
10 Initialize array
11 component_lifetime = zeros(m, n); /// Array to hold component lifetimes for each efficiency
12
13 Simulation loop
14 for j = 1:m
15     repair_efficiency = repair_efficiencies(j);
16     for i = 1:n
17         if modulo(i, initial_component_lifetime) == 0 then
18             component_lifetime(j, i) = initial_component_lifetime * repair_efficiency;
19         else
20             component_lifetime(j, i) = initial_component_lifetime;
21         end
22     end
23 end
24
25 Create 3D mesh grid
26 [Time, Efficiency] = meshgrid(time, repair_efficiencies); /// Create grid for time and repair efficiencies
27
28 Plot the 3D model
29 clf;
30 surf(Time, Efficiency, component_lifetime); /// Plotting 3D surface
31 title('3D Model of Component Lifetime with Modular Design');
32 xlabel('Time (hours)');
33 ylabel('Repair Efficiency');
34 zlabel('Component Lifetime (hours)');
```



EXPLANATION:

- The Scilab code simulates the lifetime of components over 1000 hours with varying repair efficiencies (0.7, 0.9, 1.0). It calculates how the component's lifetime changes based on periodic repairs. The results are visualized in a 3D plot where the x-axis represents time, the y-axis represents different repair efficiencies, and the z-axis shows the component lifetime. This 3D model helps visualize the impact of repair efficiency on extending the component's life over time.

An abstract geometric design on the left side of the slide. It features a dark blue background with a diagonal line running from the top-left corner towards the bottom-right. To the left of this line, there are several overlapping geometric shapes: a dark purple triangle at the top-left, a light grey semi-circle, a pattern of concentric circles, a magenta triangle with diagonal lines, a magenta triangle with horizontal lines, a magenta triangle, a grey triangle, and a magenta triangle. A small white circle is located on the diagonal line near the top-left corner.

THANK YOU

For your patience..